EL MEHDI HICHAM

Computer Vision Engineer | AI Engineer | Sports Analytics Morocco — open to relocation & remote

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PROFILE

Computer Vision Engineer specialized in **real-time video intelligence**, **multi-object tracking**, geometric vision and sports analytics. Experienced in designing and deploying end-to-end Computer Vision pipelines spanning detection, multi-object tracking, ReID, homography, OCR, pose estimation, and GPU-accelerated inference. Recent international freelance work includes football tracking-data analytics, camera calibration, Veo workflows, and technical advisory, following previous Computer Vision roles at SportScore and Priori Sport

EXPERIENCE

Computer Vision Engineer & Sports AI Consultant **07/2026 – 09/2026**
SolveXI, TechnologyOne & Multiple Global Clients — Freelance / Remote
Australia & New Zealand — Croatia

- **SolveXI / Goran Mandic:** integrated SkillCorner tracking, events and phase-of-play data into a tactical analysis pipeline producing coach-readable insights on team shape, width/depth, possession structure, passing options, pressing behaviour and cross-match patterns.
- **TechnologyOne / Eugene Luongo:** evaluated football footage quality, player detection/tracking feasibility and camera-calibration requirements; advised on Computer Vision architecture, model adaptation, Veo-camera workflows and reliable homography/pitch mapping.
- Delivered **technical advisory** on Computer Vision system design, feasibility, model selection, deployment strategy and development planning for international clients.
- Worked across **player/ball detection and tracking, homography, pitch-coordinate reconstruction, tactical analytics, model adaptation and benchmarking**.
- Translated technical requirements into clear **milestones, deliverables, implementation phases and engineering recommendations**.

Tech: Python, PyTorch, OpenCV, SkillCorner Data, Multi-Object Tracking, Homography, Tactical Analytics

Computer Vision Engineer (Contract) **05/2026 – 06/2026**
Priori Sports, Inc. — San Francisco, USA (Remote)

- Developed computer vision solutions for **football spatial tracking**, including calibrated spatial projection and tracking-data processing.
- Built **ball-height and trajectory estimation** from 2D tracking and camera-calibration data, including smoothing, airborne detection and spatial correction.

Tech: Python, OpenCV, Camera Calibration, 3D Geometry, Signal Processing, NumPy, SciPy

Computer Vision Engineer (Freelance) **09/2025 – 10/2025**
SportScore, Paris (Remote)

- Engaged after the internship to **restructure and optimize the production inference pipeline** for real-time performance.
- Refactored the pipeline architecture for parallel execution, separating model loading, pre/post-processing and inference into independent stages to remove serial bottlenecks.
- Reduced processing time by **46.7%** through CUDA Stream parallelization, TensorRT inference, batched ReID and asynchronous processing.

Tech: TensorRT FP16, CUDA Streams, ONNX, PyTorch, Python

R&D Intern — Applied AI & Computer Vision **04/2025 – 09/2025**
SportScore, Paris (Remote)

- Built an automatic frame-classification module using YOLOv11m-cls, achieving **99.95% accuracy**; implemented football-action detection including pass classification, shots, dribbles, interceptions and goals.
- Applied **camera homography and geometric transformations** for spatial analysis and pitch mapping.
- Developed player/ball detection, tracking, ReID, team assignment, jersey OCR and pose-estimation components for a football video-analysis pipeline.
- Delivered automated performance analytics including heatmaps, pass maps, structured JSON/CSV outputs and match statistics.
- Worked autonomously and contributed to coordination and code consistency across the internship team.

Tech: Python, PyTorch, TensorFlow, OpenCV, YOLO, DeepSORT, OSNet, CUDA, TensorRT, Pandas, Docker, FastAPI

PROJECTS

OCR System for Handwritten Tifinagh 2024
Hybrid CRNN (CNN + RNN) for Amazigh script recognition; benchmarked against PaddleOCR.

Autonomous Driving — Deep Reinforcement Learning 2024
ROS2 + Gazebo simulation pipeline using camera and LiDAR; TD3 agent for navigation and obstacle avoidance.

SKILLS

Computer Vision: YOLOv8/11, ByteTrack, DeepSORT, OSNet ReID, OCR, pose estimation, segmentation, homography, camera calibration, pitch mapping, 3D geometry
Sports AI & Video Analytics: player/ball tracking, tracking-data analysis, tactical analytics, pitch-coordinate reconstruction, SkillCorner data, Veo-camera workflows
Performance & Deployment: CUDA Streams, TensorRT FP16/INT8, ONNX, real-time inference, Docker, AWS (S3, EC2)
Programming: Python (advanced), C/C++, Shell, SQL, FastAPI, REST API
Deep Learning & ML: PyTorch, TensorFlow/Keras, CNN, RNN/LSTM, Transformers, EfficientNet, RegNet-Y, Bi-GRU
Mathematics: Linear algebra, projective geometry, statistical analysis, signal processing
Robotics & Tools: ROS2, Gazebo, LiDAR, Pandas, NumPy, Git, Linux
Languages: Arabic (native), French (professional), English (professional working proficiency)

EDUCATION

Master's Degree in Artificial Intelligence — *Ibn Tofail University, Kenitra* 2023 – 2025
Bachelor's Degree in Physics (Theoretical & Mathematical) — *Ibn Tofail University, Kenitra* 2020 – 2023